

Context structure and the meaning of PAST morphology

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Puzzle

Cross-linguistic tendency: **same marking** for temporal back-shifting and distance from reality (*X-marking*)

- (1) On New Year's Eve 1999, the mayor **felt** playful and **had** an idea he **would** regret one week later.
→ **temporal back shifting**
- (2) a. If I **felt** well (now), I **would** dance with you.
b. I wish I **had** a dog.
→ **distance from reality**

Proposal

- Temporal **back shifting** can be viewed as a **modal phenomenon**.
- PAST signals evaluation relative to an **incomplete** representation of the utterance **context**.

Source of variation of interpretation of PAST:

- Temporally structured** representation of the context → **back shifting**
- Temporally flat** representation of the context → **distance from reality**

Previous accounts

Past as Modal view

(e.g. Stalnaker 1968; von Fintel 1998; Iatridou 2000; Karawani 2014; Schulz 2014; Mackay 2019)

- Marking has temporal semantics in (1) but modal semantics in (2).
- Committed to form-meaning-mismatch.

⇒ **No explanation of the morphological puzzle**

Past as Past view

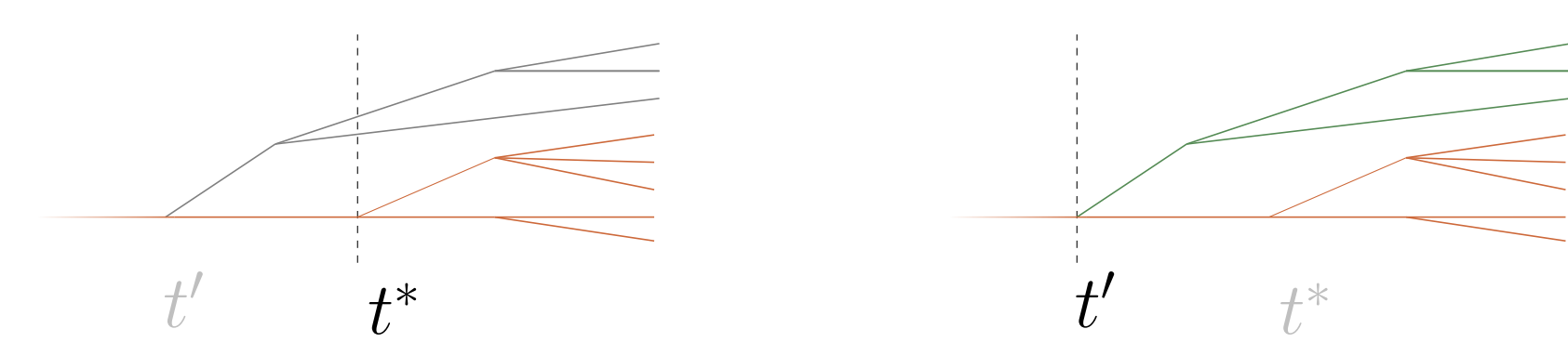
(e.g. Arregui 2009; Ippolito 2013; Ippolito & Mirrazi 2025; Khoo 2015; Romero 2014)

- PAST in (2) *also* signals temporal back shifting
- Explains the morphological puzzle

⇒ **Problematic predictions!**

Past as Past and its challenges

PAST in distancing from reality signals **temporal back shifting** of the point from which historical possibilities are accessed. (e.g. Arregui 2009; Ippolito 2013; Ippolito & Mirrazi 2025; Khoo 2015; Romero 2014)



- (3) If I **felt** well (now), I **would** dance with you.
≈ *Among the options that were still possible at t', those in which I feel well at t*, are st. I dance with you at t*.*
≈ It was the case that: If I feel better at t*, I dance with you at t*

Problematic prediction 1

- Temporal back shifting introduces a time that can be picked up with *back then*.
 - This is not the case with PAST in distance from reality constructions.
- (4) She **went** to Stanford.
... She had red hair back then.
- (5) If I **felt** well, I **would** dance with you.
... #I hadn't eaten the raw meat dish back then.

Problematic prediction 2

- PAST morphology should not occur in sentences about scenarios that *never were* possible.
 - However, PAST morphology is **obligatory** in counter-mathematics. (cf. Lewis 1973, Mizuno & Kaufmann 2019)
- (6) If two plus two **was** ten, a lot of things **would** be different.

Conclusion

- Distancing from reality is not a temporal phenomenon. PAST morphology in these cases does not signal a temporal back shift.
- Does not mean we have to give up unified analysis of PAST morphology in temporal back shifting and distancing from reality.
- Instead: **both** can be understood as **modal phenomena**.

Proposal

Gist of proposal

- Two ways to represent** an utterance **context**:
⇒ **Temporally structured**
⇒ **Temporally flat**
- PAST signals: **hold fixed only a subset** of the maximal set of contextually settled facts.

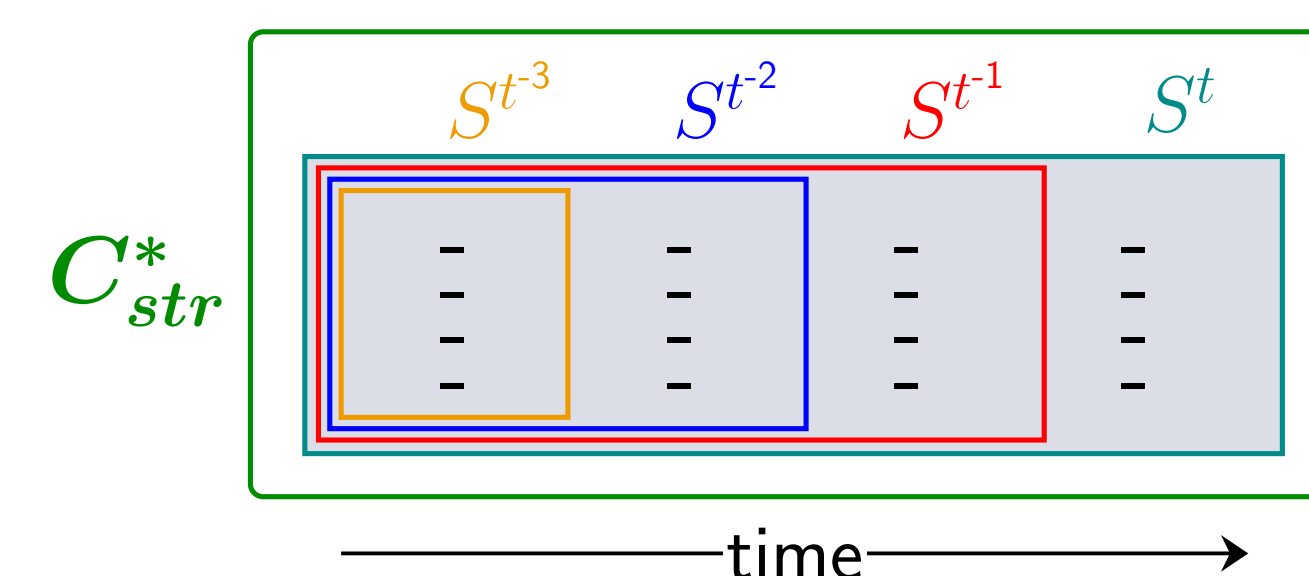
Temporally structured context set C_{str}^*

- Temporally structured context set C_{str}^* : the set containing for all times t not later than t^* the temporal state S^{t,w^*}

Defined relative to indices w^*, t^* : C_{str}^* is the set of temporal states st. $\forall t \leq t^* : S^{t,w^*} \in C_{str}^*$

- Temporal states $S^{t,w}$: the set of all propositions metaphysically settled relative to w and t

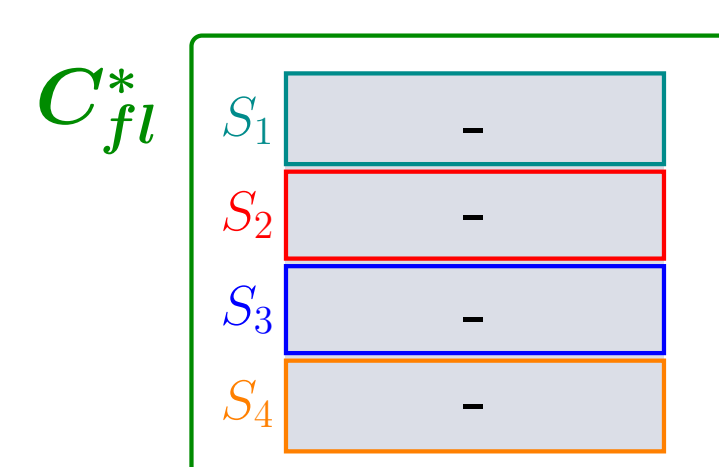
For any time t and world w ,
the temporal state $S^{t,w}$ is the set of propositions $p_{(s,t)}$ st.
 $p \in S^{t,w}$ iff $\forall w' [w \simeq_t w' \rightarrow p(w') = 1]$



Temporally flat context set C_{fl}^*

- The set containing for each proposition p settled at t^* the singleton set $\{p\}$. Alternatively: causally structured set (Kaufmann 2013; Santorio 2019).

Defined relative to indices w^*, t^* , C_{fl}^* is the set of (singleton) sets S of propositions st: $\forall p [\forall w [w^* \simeq_{t^*} w \rightarrow p(w) = 1] \rightarrow \{p\} \in C_{fl}^*]$



Evaluation relative to factive sets

Interpretation function defined relative to two factive sets

- Context set C^*
 - Metaphysical or epistemic representation of (assumed) reality
 - Can be temporally structured or flat
- Evaluation set V
 - V stands in a certain relation to C^*
 - Certain linguistic operators specify this relation (e.g. tense)

$$\llbracket \bullet \rrbracket^{g,w,t,C^*,V}$$

Truth of a sentence

- A sentence ϕ is true relative to g, w, t, C^*, V iff ϕ is true throughout all worlds compatible with V (cf. Yalcin's (2007) domain semantics)

$$\llbracket \phi \rrbracket^{g,w,t,C^*,V} = 1 \text{ iff } \forall w' \in \bigcup V : \llbracket \phi \rrbracket^{g,w',t,C^*,V} = 1$$

The meaning of PAST

- PAST morphology realizes abstract feature SUB.
- SUB signals evaluation relative to a non-maximal set of available facts.
- Presupposes that evaluation set V is a **less informative** contextually salient **subset** of set C^* (cf. Mackay 2019).

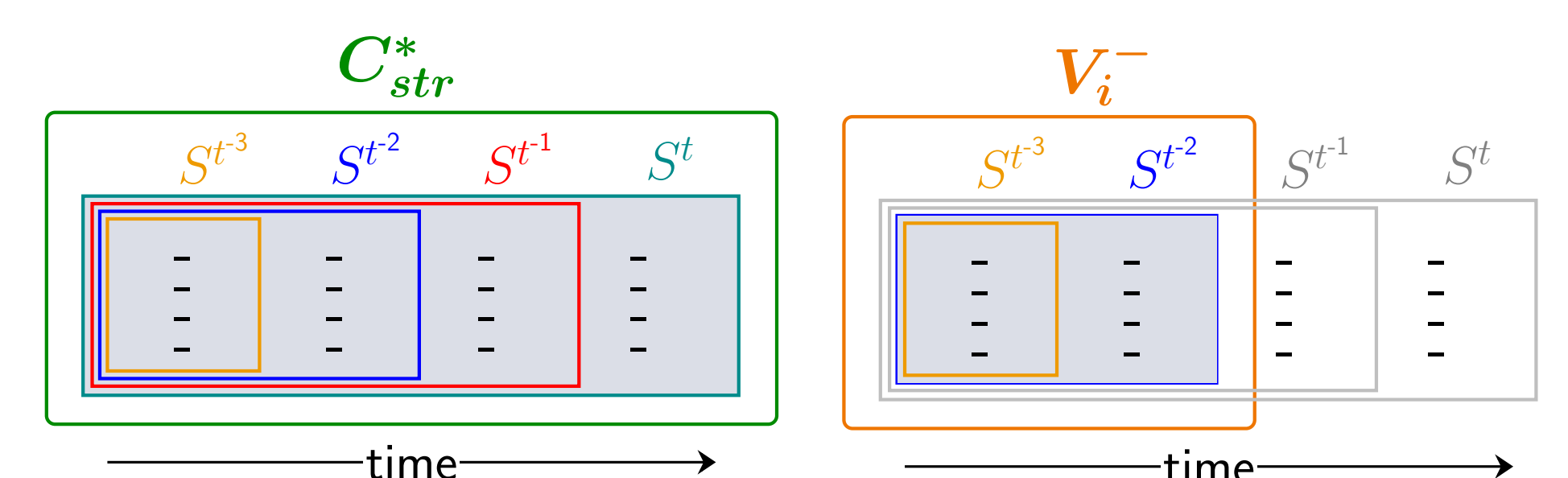
$$\llbracket \text{SUB}_i \phi \rrbracket^{g,w,t,C^*,V} \text{ is defined only if } \llbracket \phi \rrbracket^{g,w,t,C^*,V_i^-} = 1$$

$$\text{where } V_i^- \subset C^* \text{ and } \bigcup V_i^- \subset \bigcup C^*$$

The effect of PAST

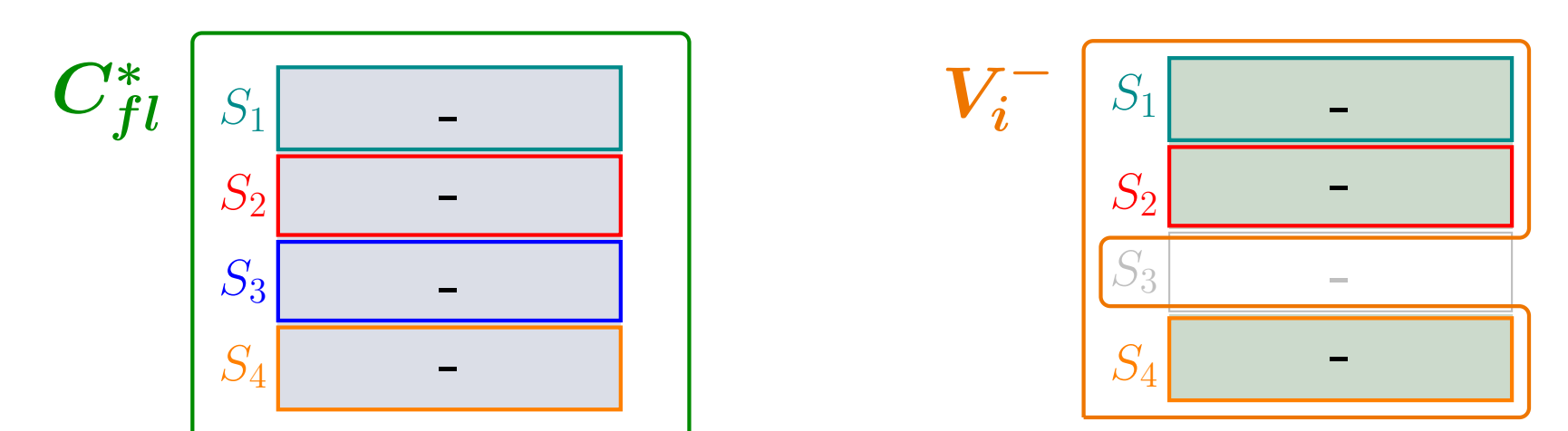
Temporally structured context set C_{str}^*

- Evaluation at a less informative subset of C_{str}^* → evaluation at a **previous stage of history**
- Truth of a sentence: ϕ has to hold at *all* worlds sharing previous stage of history (branching future) → temporal back shifting of ϕ



Temporally flat context set C_{fl}^*

- Less informative subset of C_{fl}^* : evaluation relative to factive set with certain facts suspended → abstracting away from certain aspects of reality
- Domain expansion/presupposition suspension view of X-marking (e.g. Stalnaker 1968; von Fintel 1998; Iatridou 2000; Karawani 2014; Schulz 2014; Mackay 2019)



Conclusion

- Temporal shifting can be conceived in modal terms
- Allows for a unified analysis of back shifting and distancing from reality
- Variation in interpretation of PAST: flexibility in our conception of context
- Reconciliation of Past as Past and Past as Modal views

References

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